SHEAFIFICATION PRESERVES LIMIT PRESERVING FUNCTORS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 512-594, following the statement at 505-510.

Editorial note. No omissions; exactness of filtered colimits and the colimit commutation are anchored in the text.

Allowed standard prerequisites (editorial).

- Convergence of the plus construction. Sites, Theorem sites-theorem-plus (cross-chapter, cited at ~515-516). F^# = (F^+)^+, so it suffices to show F^+ is limit preserving.
- Standard fppf coverings are cofinal. Topologies, Lemma topologies-lemma-fppf-affine (cross-chapter, cited at ~521-523). The colimit defining F^+(T) may be taken over standard coverings only.
- Schemes over limits. Limits, Lemma limits-lemma-scheme-over-limit (cross-chapter, cited at ~577-579). lim T_{i',k} = T_k and the analogous fibre-product identity.
- Colimits commute with colimits. Categories, Lemma categories-lemma-colimits-commute (cross-chapter, cited at ~592-593). Justifies switching the order of the two colimits.

Source reference. Original reference 002M: focused-seeds/geometry/S9-spaces-limits-sheafify-finite-presentation/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01YZ: focused-seeds/geometry/S9-spaces-limits-sheafify-finite-presentation/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00WB: focused-seeds/geometry/S9-spaces-limits-sheafify-finite-presentation/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 021P: focused-seeds/geometry/S9-spaces-limits-sheafify-finite-presentation/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction

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the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme contained in Sch_{fppf}. Let $F : (Sch/S)^{opp}_{fppf} \to Sets$ be a functor. If F is limit preserving then its sheafification $F^\#$ is limit preserving.*

Proof. Assume F is limit preserving. It suffices to show that F^+ is limit preserving, since $F^\# = (F^+)^+$, see Sites, Theorem 00WB. Let T be an affine scheme over S, and let $T = \lim T_i$ be written as the directed limit of an inverse system of affine S schemes. Recall that $F^+(T)$ is the colimit of $\check{H}^0(\mathcal{V}, F)$ where the limit is over all coverings of T in $(Sch/S)_{fppf}$. Any fppf covering of an affine scheme can be refined by a standard fppf covering, see Topologies, Lemma 021P. Hence we can write

$$F^+(T) = \operatorname{colim}_{\mathcal{V} \text{ standard covering } T} \check{H}^0(\mathcal{V}, F).$$

Any $\mathcal{V} = \{T_k \to T\}_{k=1,\ldots,n}$ in the colimit may be written as $\mathcal{V}_i \times_{T_i} T$ for some i and some standard fppf covering $\mathcal{V}_i = \{T_{i,k} \to T_i\}_{k=1,\ldots,n}$ of T_i. Denote $\mathcal{V}_{i'} = \{T_{i',k} \to T_{i'}\}_{k=1,\ldots,n}$ the base change for $i' \geq i$. Then we see that

$$\begin{aligned}
\operatorname{colim}_{i' \geq i} \check{H}^0(\mathcal{V}_i, F) &= \operatorname{colim}_{i' \geq i} \text{Equalizer}\Big(\prod F(T_{i',k}) \rightrightarrows \prod F(T_{i',k} \times_{T_{i'}} T_{i',l}) \Big) \\
&= \text{Equalizer}\Big(\operatorname{colim}_{i' \geq i} \prod F(T_{i',k}) \rightrightarrows \operatorname{colim}_{k' \geq k} \prod F(T_{i',k} \times_{T_{i'}} T_{i',l}) \Big) \\
&= \text{Equalizer}\Big(\prod F(T_k) \rightrightarrows \prod F(T_k \times_T T_l) \Big) \\
&= \check{H}^0(\mathcal{V}, F)
\end{aligned}$$

Here the second equality holds because filtered colimits are exact. The third equality holds because F is limit preserving and because $\lim_{i' \geq i} T_{i',k} = T_k$ and $\lim_{i' \geq i} T_{i',k} \times_{T_{i'}} T_{i',l} = T_k \times_T T_l$ by Limits, Lemma 01YZ. If we use this for all coverings at the same time we obtain

$$\begin{aligned}
F^+(T) &= \operatorname{colim}_{\mathcal{V} \text{ standard covering } T} \check{H}^0(\mathcal{V}, F) \\
&= \operatorname{colim}_{i \in I} \operatorname{colim}_{\mathcal{V}_i \text{ standard covering } T_i} \check{H}^0(T \times_{T_i} \mathcal{V}_i, F) \\
&= \operatorname{colim}_{i \in I} F^+(T_i)
\end{aligned}$$

The switch of the order of the colimits is allowed by Categories, Lemma 002M. □